

Statistical Comparison of Model Performance using k-Fold Cross-Validation and Wilcoxon Test

Fernando Pujaico Rivera



A performs better than B?

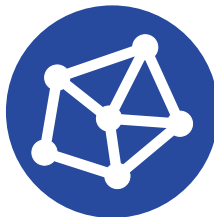
Architecture

A



Architecture

B



VS

Dataset



?

Problem

Goal: determine whether **Architecture A** performs better than **Architecture B**.

Method: **k-fold cross-validation + Wilcoxon signed-rank test**

Objective:

- Obtain multiple performance estimates
- Perform a **statistical comparison between architectures**

k-Fold Cross-Validation

Dataset is divided into K folds. If $K = 5$, then

fold1 fold2 fold3 fold4 fold5

Iteration 1 **val** **train** **train** **train** **train** ➔ **score 1**

k-Fold Cross-Validation

Dataset is divided into K folds. If $K = 5$, then

	fold1	fold2	fold3	fold4	fold5	
Iteration 1	val	train	train	train	train	→ score 1
Iteration 2	train	val	train	train	train	score 2
Iteration 3	train	train	val	train	train	score 3
Iteration 4	train	train	train	val	train	score 4
Iteration 5	train	train	train	train	val	score 5

K-Fold Cross-Validation

Example with $K = 5$.

Validation fold	A Score A	B Score B
1	a_1	b_1
2	a_2	b_2
3	a_3	b_3
4	a_4	b_4
5	a_5	b_5

$$\mu_A = \frac{1}{K} \sum_{k=1}^K a_k \quad \mu_B = \frac{1}{K} \sum_{k=1}^K b_k$$

Paired Comparison

Because both models are evaluated on the same folds, the comparison is **paired**.

For each fold:

$$d_k = a_k - b_k$$

We analyze the distribution of:

$$\mathbf{d} = [d_1, d_2, d_3, d_4, d_5]$$

Key idea: We test whether the distribution of d_k is centered above zero.

Wilcoxon signed-rank test

What it tests:

- Whether the distribution of paired differences $d_k = a_k - b_k$
- is centered above zero

Why not a t-test?

- Does not assume normality
- Works with small sample sizes
- Based on ranks, not raw means

Hypothesis Test

We test whether model A tends to outperform model B.

Define:

$$d_k = a_k - b_k$$

$$H_0 : \text{median}(\mathbf{d}) \leq 0 \quad H_1 : \text{median}(\mathbf{d}) > 0$$

Interpretation: If $p < 0.05$, A tends to outperform B on the same folds.

Wilcoxon Test in Python



alternative="greater" $\Rightarrow$ tests whether $d_k = A - B > 0$

```
import numpy as np
from scipy import stats

A = np.array([0.82, 0.79, 0.83, 0.81, 0.80])
B = np.array([0.80, 0.76, 0.81, 0.78, 0.78])

stat, p = stats.wilcoxon(A, B, alternative="greater")

if p < 0.05:
    print("A performs significantly better than B")
```

Important Limitation

Note:

- Fold results are **not independent**
- Training sets overlap across folds
- Therefore, statistical tests on k-fold are **approximations**

Implication:

- Results should be interpreted with caution

Conclusion

Procedure:

- 1 Train models using **k-fold cross-validation**
- 2 Obtain paired scores (a_k, b_k)
- 3 Compute differences $d_k = a_k - b_k$
- 4 Apply a **one-sided Wilcoxon signed-rank test**

if $p < 0.05$ **then**

 | We conclude that A tends to outperform B;

else

 | No significant difference;

Does A really perform better than B?

Now we have a principled way to answer that.

Thank you for watching!